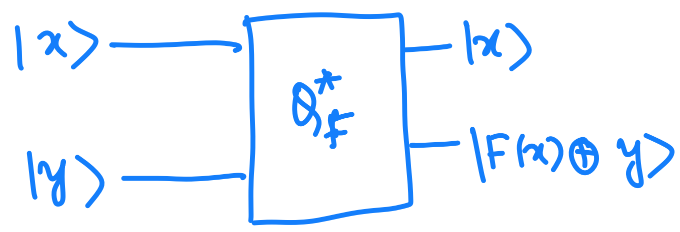
LECTURE 6

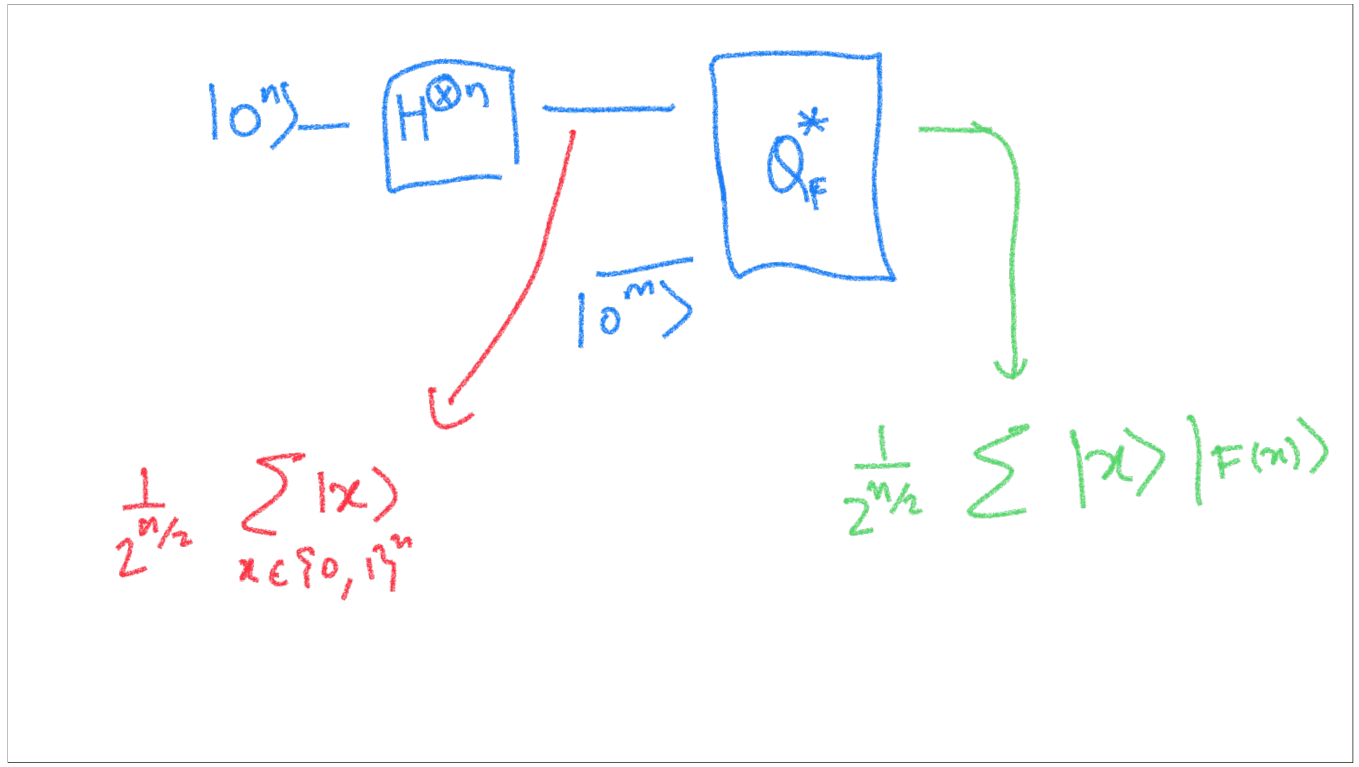
If there is a efficient classical algo/circuit that computes $F : \{0,1\}^n \to \{0,1\}^m$



$y = 0^m$ gives $|F(x)\rangle$

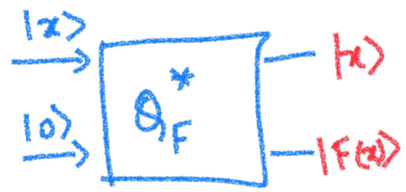
$y \in \{0,1\}^m$

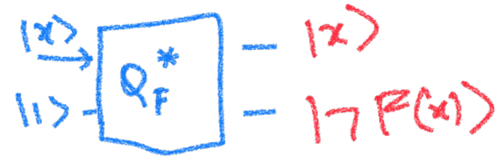
We will not always write the ancilla qubits $|0^a\rangle$

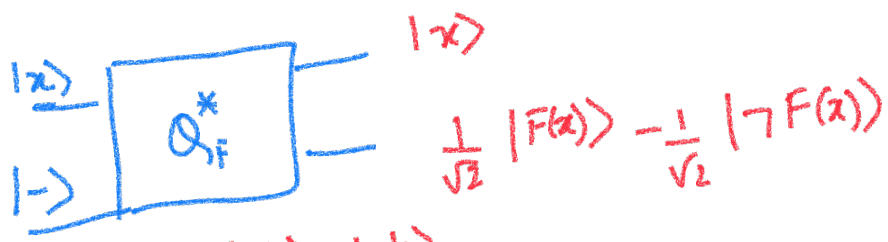


$|0^n\rangle$
$H^{\otimes n}$
Q_F^*
$|0^m\rangle$
$\frac{1}{2^{n/2}} \sum_{x \in \{0,1\}^n} |x\rangle$
$\frac{1}{2^{n/2}} \sum |x\rangle |F(x)\rangle$

For $F : \{0,1\}^n \to \{0,1\}$







$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$

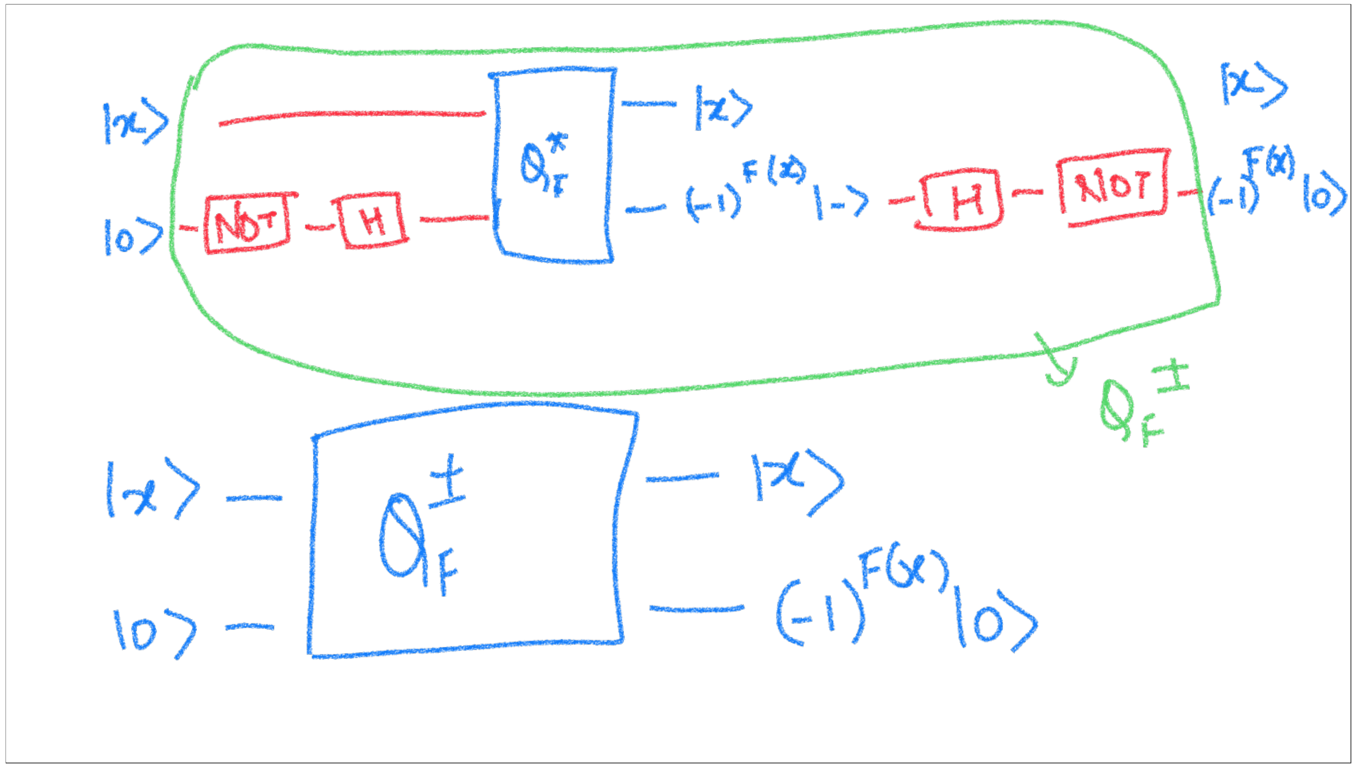
$F(x) = 0$

we get $|x\rangle \otimes |-\rangle$

$F(x) = 1$

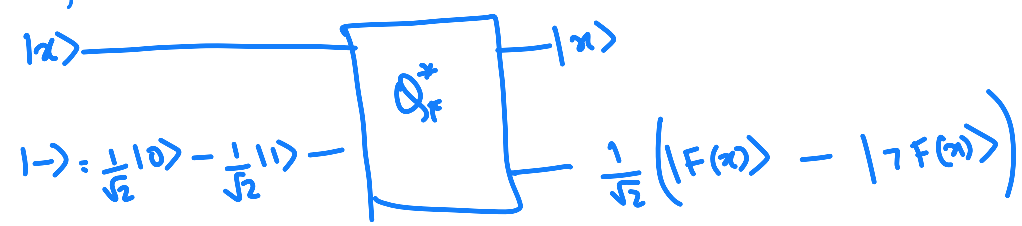
$-|x\rangle \otimes \quad |-\rangle$

$(-1)^{F(x)} \quad |x\rangle \otimes |-\rangle$



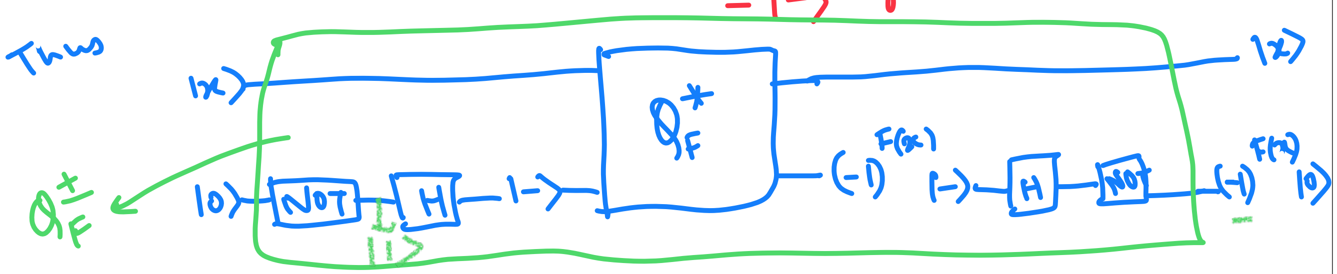
|x⟩
|0⟩
NOT
H
Q_F^*
|x⟩
$(-1)^{F(x)}$ |-⟩
H
NOT
|x⟩
$(-1)^{F(x)}$ |0⟩
$Q_F^{\pm}$
|x⟩
|0⟩
$Q_F^{\pm}$
|x⟩
$(-1)^{F(x)}$ |0⟩

Useful trick : $Q_F^{\pm}$ for $F : \{0,1\}^n \to \{0,1\}$



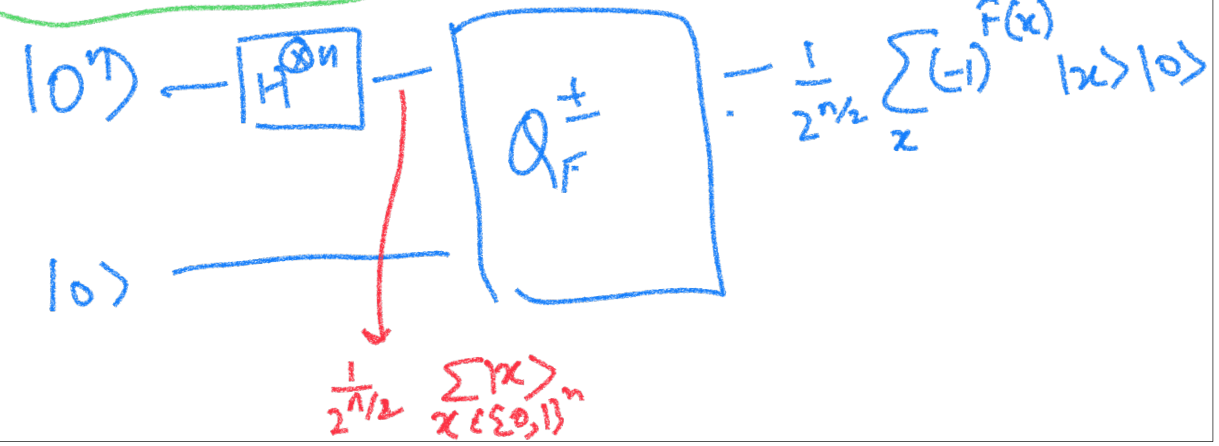
$$\left.\begin{array}{ll} |-\rangle & \text{if } F(x) = 0 \\ -|-\rangle & \text{if } F(x) = 1 \end{array}\right\} (-1)^{F(x)}|-\rangle$$

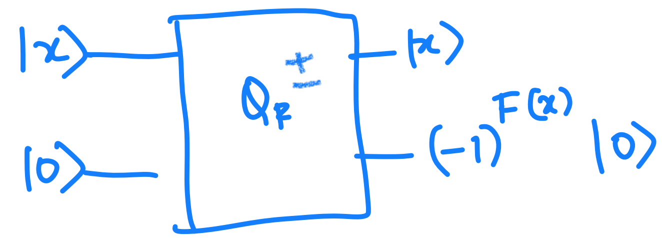
Thus



What if we want

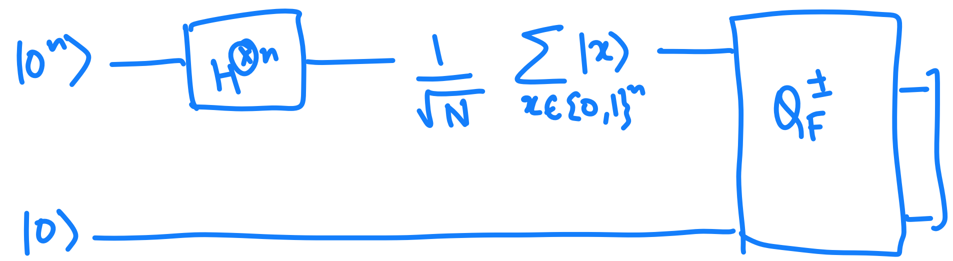
$$\boxed{\frac{1}{2^{n/2}} \sum_{x \in \{0,1\}^n} (-1)^{F(x)} |x\rangle}$$





$(-1)^{F(x)}\;|x\rangle \otimes |0\rangle$

Useful idea (A single application of $Q_F^{\pm}$ creates a state which has info about $F(x)$ for all x.)



$$\frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle \otimes \left((-1)^{F(x)} |0\rangle \right)$$

$$= \frac{1}{\sqrt{N}} \sum_{x} (-1)^{F(x)} |x\rangle \otimes |0\rangle$$

$N = 2^n.$

Consider the state $\frac{1}{2^{n/2}} \sum_{x \in \{0,1\}^n} (-1)^{F(x)} |x\rangle$

$$\frac{1}{2^{n/2}} \begin{array}{cl} \begin{bmatrix} 1 \\ -1 \\ -1 \\ \vdots \\ (-1)^{F(x)} \\ \vdots \\ -1 \end{bmatrix} & \begin{array}{l} \\ \\ \\ \\ \leftarrow |x\rangle \\ \\ \\ \end{array} \end{array}$$

More generally $g: \{0,1\}^n \to \mathbb{C}$ can be interpreted as a quantum state if

$$\frac{1}{N} \sum_{x \in \{0,1\}^n} |g(x)|^2 = 1$$

Given $g : \{0,1\}^n \to \mathbb{C}$ with $\frac{1}{N} \sum_{x \in \{0,1\}^n} |g(x)|^2 = 1$

$$|g\rangle = \frac{1}{\sqrt{N}} \begin{bmatrix} g(0\ldots0) \\ \vdots \\ \vdots \\ \vdots \\ g(11\ldots1) \end{bmatrix}$$

$N = 2^n$.

$$\frac{|g(0\ldots0)|^2}{N} + \frac{|g(0\ldots1)|^2}{N} + \cdots + \frac{|g(1\ldots1)|^2}{N} = 1$$

- So how do we use the power of quantum?
- Is there more that we can do with quantum gates?
- We will assume we can efficiently implement any 1/2/3 qubit transformation.

We want transformations that are

- easy to implement via simple gates/transformations
- Useful.

Q.C research is an interplay between these two goals.

- Sometimes we will get easy transformations and see how useful these are.

- With more experience, we will do a bit of both.

Possible simple transformations

$|x_1 \ldots x_n\rangle$ — $\boxed{H^{\otimes n}}$?

Fourier transform over $\{0,1\}^n$ or $\mathbb{Z}_2^n$

$$|0\rangle \longrightarrow \boxed{H} — |+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|1\rangle \longrightarrow \boxed{H} — |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

For $s \in \{0,1\}$

$$|s\rangle — \boxed{H} — \frac{1}{\sqrt{2}}|0\rangle + (-1)^{s}\frac{1}{\sqrt{2}}|1\rangle$$

$$= \frac{1}{\sqrt{2}}\sum_{x \in \{0,1\}} (-1)^{x \cdot s}|x\rangle$$

$$|s_1 s_2\rangle \;—\; \boxed{H \otimes H} \qquad |Hs_1\rangle \otimes |Hs_2\rangle$$

$$= \frac{1}{2} \sum_{x_1 \in \{0,1\}} (-1)^{x_1 s_1} |x_1\rangle \otimes \sum_{x_2 \in \{0,1\}} (-1)^{x_2 s_2} |x_2\rangle$$

$$= \frac{1}{2} \sum_{\substack{x_1 \in \{0,1\} \\ x_2 \in \{0,1\}}} (-1)^{x_1 s_1} \cdot (-1)^{x_2 s_2} |x_1 x_2\rangle$$

$$= \frac{1}{2} \sum_{x_1, x_2} (-1)^{x_1 s_1 + x_2 s_2} |x_1 x_2\rangle$$

$$|s_1 s_2 s_3\rangle = \frac{1}{\sqrt{8}} \sum_{x_1, x_2, x_3 \in \{0,1\}} (-1)^{x_1 s_1 + x_2 s_2 + x_3 s_3} |x_1 x_2 x_3\rangle$$

$$|s_1 s_2 \ldots s_n\rangle \longrightarrow \boxed{H^{\otimes n}} \longrightarrow |Hs_1\rangle \otimes \ldots |Hs_n\rangle$$

$$\downarrow$$

$$\left(\frac{1}{\sqrt{2}}\right)^n \sum_{x_1,\ldots,x_n \in \{0,1\}} (-1)^{x_1 s_1 + \ldots + x_n s_n} |x_1 \ldots x_n\rangle$$

- We now know what $H^{\otimes n}$ does to base states.

- What about superposition of base states.

Use linearity

We call

$$\frac{1}{2^{n/2}} \sum_{\vec{x} \in \{0,1\}^n} (-1)^{\sum x_i s_i} |x\rangle \quad \text{as } \chi_{\vec{s}}$$

$$|\vec{s}\rangle \; — \; \boxed{H^{\otimes n}} \; — \; |\chi_s\rangle$$

$$\sum \alpha_s \vec{s} \; — \; \boxed{H^{\otimes n}} \; — \; \sum \alpha_s |\chi_s\rangle$$

$$|x\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{array}{l} 000 \\ 001 \\ 010 \\ 011 \\ 100 \\ 101 \\ 110 \\ 111 \end{array} \longrightarrow \frac{1}{\sqrt{8}} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ -1 \\ 1 \\ -1 \\ 1 \end{bmatrix} \rightarrow |\chi_x\rangle$$

$\rightarrow s^{th}$ co-ordinate $(-1)^{\sum x_i s_i \pmod 2}$

More generally

$$\boxed{x} - \boxed{H^{\otimes n}} - \boxed{\chi_x}$$

$$\chi_x = \begin{bmatrix} \\ \\ \\ \\ \end{bmatrix}$$

$\vec{s}$

$\vec{s}^{th}$ co-ordinate $(-1)^{\sum_{i=1}^{n} x_i s_i \pmod 2}$

$= |\chi_{\vec{x}}\rangle$

The transformation that

$$H^{\otimes n} = \begin{bmatrix} | & | & & | \\ x_{0\ldots0} & x_{0\ldots01} & \cdots & x_{11\ldots1} \\ | & | & & | \end{bmatrix}_{2^n \times 2^n}$$

$$H^{\otimes n} \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |X_x\rangle \xrightarrow{H^{\otimes n}} \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle$$

$$(H^{\otimes n})^{-1} = H^{\otimes n}$$

Key properties

$$\langle \chi_{\vec{s}} | \chi_{\vec{t}} \rangle = \langle t | s \rangle = 1 \text{ if } \vec{t} = \vec{s}$$
$$= 0 \text{ otherwise}$$

$$|x\rangle \xrightarrow{H^{\otimes n}} |\chi_x\rangle \xrightarrow{H^{\otimes n}} |x\rangle$$

Q Given n qubit state $|\psi\rangle$, $|\psi\rangle = \begin{bmatrix} \alpha_{00\ldots0} \\ \vdots \\ \vdots \\ \alpha_{111\ldots1} \end{bmatrix}$

$$|\psi\rangle \xrightarrow{H^{\otimes n}} |\phi\rangle$$

$$|\phi\rangle = \sum_{\vec{v} \in \{0,1\}^n} \beta_{\vec{v}} |\vec{v}\rangle .$$

What is $\beta_{\vec{v}}$?

$$\langle v | \phi\rangle = \langle \chi_v | \psi\rangle$$

$$|\psi\rangle = \sum_{\vec{v} \in \{0,1\}^n} \alpha_{\vec{v}} |\vec{v}\rangle = \underbrace{\sum \beta_v |\chi_v\rangle}$$

$$|\phi\rangle = H^{\otimes n} |\psi\rangle = \sum \alpha_{\vec{v}} |\chi_v\rangle = \sum \underline{\beta_v} |v\rangle$$

$$\beta_v = \langle \chi_v | \psi \rangle$$

Recall that a function can be written as a quantum state

$$g : \{0,1\}^n \longrightarrow \mathbb{C}$$

Assumption : $\frac{1}{N} \sum_{x \in \{0,1\}^n} |g(x)|^2 = 1$

$$|g\rangle = \frac{1}{\sqrt{N}} \begin{bmatrix} g(0\ldots 0) \\ g(0\ldots 01) \\ \vdots \\ g(1\ldots 1) \end{bmatrix} = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} g(x)\, |x\rangle$$

Example $F : \{0,1\}^n \to \{0,1\}$

$$f(x) = (-1)^{F(x)} \in \{-1, 1\}$$

$$|f\rangle = \frac{1}{2^{n/2}} \begin{bmatrix} +1 \\ -1 \\ \vdots \\ (-1)^{F(x)} \\ \vdots \end{bmatrix} \to \vec{x}\text{'th co-ordinate}$$

2^{2^n} such functions

Consider $g : \{0,1\}^n \to \mathbb{C}$

$$|g\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} g(x) |x\rangle = \sum_{s \in \{0,1\}^n} \hat{g}(s) |\chi_s\rangle$$

Fourier coeff. on s

$$\hat{g}(s) = \langle \chi_s | g \rangle$$

$$|g\rangle - \boxed{H^{\otimes n}} \to \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} g(x) |\chi_x\rangle$$

$$= \sum_{s \in \{0,1\}^n} \hat{g}(s) |s\rangle$$

$$|g\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} g(x)\, |x\rangle = \sum_{x \in \{0,1\}^n} \hat{g}(x)\, |\chi_x\rangle$$

$$\sum |\hat{g}(x)|^2 = 1$$

$\forall x$

$$\hat{g}(x) = \langle \chi_x | g \rangle = \frac{1}{N} \sum_{s \in \{0,1\}^n} \chi_x(s)\, g(s)$$

If $g(x) = \pm 1$ for all x, then

$$\hat{g}(x) = \frac{\#\{s : \chi_x(s) = g(s)\}}{N} - \frac{\#\{s : \chi_x(s) = -g(s)\}}{N}$$

When is $\hat{g}(s)$ large (By large, we mean absolute value close to 1)

If $g(x)$ and $\chi_s(x)$ agree for many x

Intuitively, this means $g(x)$ "looks like" $\chi_s(x)$

Deutsch Jozsa :

Let $F : \{0,1\}^n \to \{0,1\}$ s.t. F is either constant (all 0 or all 1)

or F is balanced.

Question : Decide whether Constant/balanced.

Assume : Given a circuit to compute F.

# of queries to F

Deterministic : $2^{n-1}+1$

Randomized : Probability p success will require $\log \frac{1}{1-p}$

Quantum :

F is const.

$$\frac{1}{\sqrt{N}}\begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \qquad \frac{1}{\sqrt{N}}\begin{bmatrix} -1 \\ \vdots \\ -1 \\ -1 \\ -1 \\ -1 \\ \vdots \\ -1 \end{bmatrix}$$

$$\chi_{0\ldots0} \qquad\qquad -\chi_{0\ldots0}$$

F is balanced

$$f(x) := (-1)^{F(x)}$$

$$|f\rangle = \frac{1}{\sqrt{N}}\begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \\ \vdots \\ 1 \end{bmatrix} = \sum \hat{f}(s)|\chi_s\rangle$$

$$\Big\downarrow H^{\otimes n}$$

$$\sum_{s\in\{0,1\}^n} \hat{f}(s)\,|s\rangle$$

$$\hat{f}(0\ldots0) = \langle \chi_{0\ldots0}, f\rangle = \frac{1}{N}\left(\frac{N}{2} - \frac{N}{2}\right) = 0$$

$$0^n \xrightarrow{H^{\otimes n}} \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle \longrightarrow \boxed{O_F^{\pm}} \longrightarrow \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} (-1)^{F(x)} |x\rangle$$

$N = 2^n$.

$$f(x) = (-1)^{F(x)}$$

We have $|f\rangle = \frac{1}{\sqrt{N}} \sum f(x) |x\rangle$

$$|f\rangle \xrightarrow{H^{\otimes n}} \boxed{\text{Measure}}$$

→ If f is const. then $|0^n\rangle$ w.p. 1

→ If f is balanced then $|0^n\rangle$ w.p. 0.

$$|0^n\rangle \;—\; \boxed{H^{\otimes n}} \;—\; \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle \;—\; \boxed{O_F^{\pm}} \;—\; \frac{1}{\sqrt{N}} \sum_{x} (-1)^{F(x)} |x\rangle$$

$$\frac{1}{\sqrt{N}} \sum_{x} (-1)^{F(x)} |x\rangle = |f\rangle$$

$$f(x) = (-1)^{F(x)} \qquad f : \{0,1\}^n \to \{-1, 1\}$$

$$\hat{f}(x) = \frac{\#\{s : \chi_x(s) = f(s)\}}{N} - \frac{\#\{s : \chi_x(s) = -f(s)\}}{N}$$

$\hat{f}(0^n) = 1$ if $f \equiv 1$,

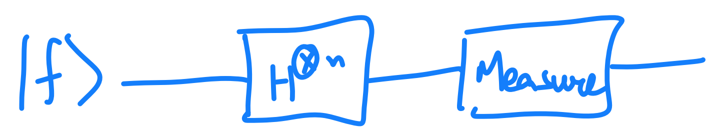
$= -1$ if $f \equiv -1$

$\hat{f}(0^n) = 0$ if f is balanced.

Thus

$|f\rangle = \pm|x_{0^n}\rangle$ if f is constant

and the co-efficient of $|x_{0^n}\rangle$ is 0, if f is balanced.



$|0^n\rangle$ if f is constant

$y \neq 0^n$ if f is balanced.

Bernstein Vazirani :

Let: $F : \{0,1\}^n \to \{0,1\}$

$$F(x) = \sum_{i=1}^{n} x_i s_i \pmod 2$$ for some

unknown $s \in \{0,1\}^n$.

Find s.

$$0^n \xrightarrow{H^{\otimes n}} \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle \longrightarrow \boxed{O_F^{\pm}} \longrightarrow \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} (-1)^{F(x)} |x\rangle$$

$$f(x) = (-1)^{F(x)}$$

$N = 2^n$.

We have $|f\rangle = \frac{1}{\sqrt{N}} \sum f(x) |x\rangle$

$$= \frac{1}{\sqrt{N}} \sum_x (-1)^{\sum x_i s_i \pmod 2} |x\rangle$$

$$= |\chi_s\rangle$$

$$|\chi_s\rangle \xrightarrow{H^{\otimes n}} |\vec{S}\rangle \longrightarrow \boxed{\text{Measure}}$$

Simon's algorithm:

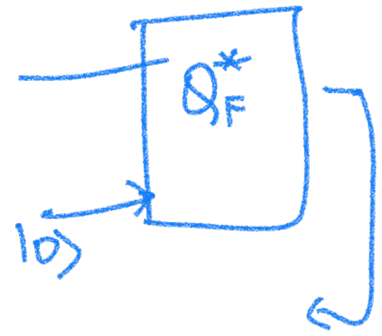
Given $F : \{0,1\}^n \longrightarrow \{0,1\}^m$ s.t. $\exists\, L \in \{0,1\}^n$

that satisfies

$\forall x, y \in \{0,1\}^n$ $\quad F(x) = F(y) \iff y = x \oplus L.$

Goal: Find L.

$$0^n \longrightarrow H^{\otimes n} \longrightarrow \frac{1}{2^{n/2}} \sum_{x \in \{0,1\}^n} |x\rangle$$



$$\frac{1}{2^{n/2}} \sum_{x \in \{0,1\}^{n/2}} |x\rangle \, |F(x)\rangle$$

If we measure the last m qubits, we will see one of $2^m/2$ outputs.

State collapses to $\frac{1}{\sqrt{2}} \left(|y\rangle + |y \oplus L\rangle \right) \otimes$ ......

For a random $y \in \{0,1\}^n$

If we repeat

$$|y_1\rangle + |y_1 \oplus L\rangle$$

$$|y_2\rangle + |y_2 \oplus L\rangle$$

$$\vdots$$

Solution :

Step 1 : Create $\frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle \otimes |F(x)\rangle$, and measure last n qubits

State collapses to $\frac{1}{\sqrt{2}} \left(|y\rangle + |y \oplus L\rangle \right) \otimes |F(y)\rangle$

for some y.

Step 2: Hadamard on the 1st n qubits.

State changes to

$$\frac{1}{\sqrt{2}}\left(|x_y\rangle + |x_{y\oplus L}\rangle\right) \otimes |F(y)\rangle$$

s^{th} co-ordinate

$$(-1)^{\sum y_i s_i \pmod 2} + (-1)^{\sum y_i (L_i + s_i) \pmod 2}$$

$$= (-1)^{\sum y_i s_i}\left[(-1)^{\sum L_i y_i} + 1\right] = 0 \text{ if } \sum L_i y_i \text{ is odd.}$$

Step 3: Measure the 1st n qubits

We get a random y s.t. $\sum y_i L_i = 0 \pmod 2$

Repeat $O(n)$ times to get linear equations in $L_1, \ldots, L_n$. Solve to get $L = L_1, \ldots, L_n$.